Use Case Diagrams

17-626 Requirements for Information Systems

Prof. Travis Breaux

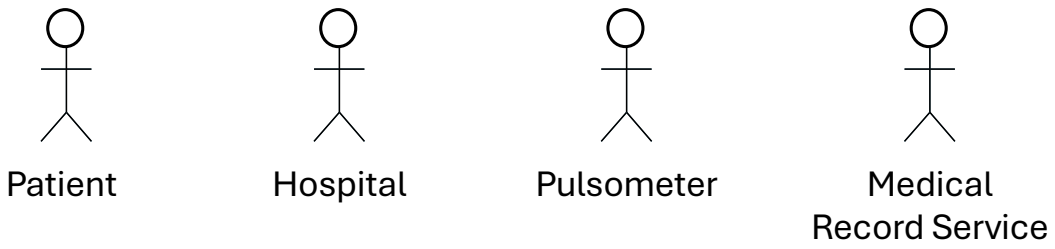
Basic Use Cases

Use Cases help requirements analysts to…

- Identify actors and events around the system
- Define the system boundary – what is or is not within the system scope?
- Investigate early design interactions
  (uses cases need not be descriptions of the final design)

Defining Actors

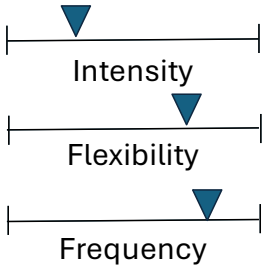
- An actor is an entity that interacts with the system for the purpose of completing an event [Jacobson, 1992]
- Actors can be a user, an organization, a device, or an external system



Personas to represent stakeholders



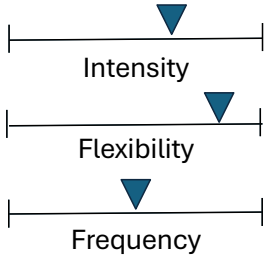
Name: Patrick
Age: 72
Gender: Male
Job: Retired Financial Analyst



Patrick is an information seeker, and believes in frequent exercise but is concerned about exceeding his limits. He lacks confidence about intense exercise.

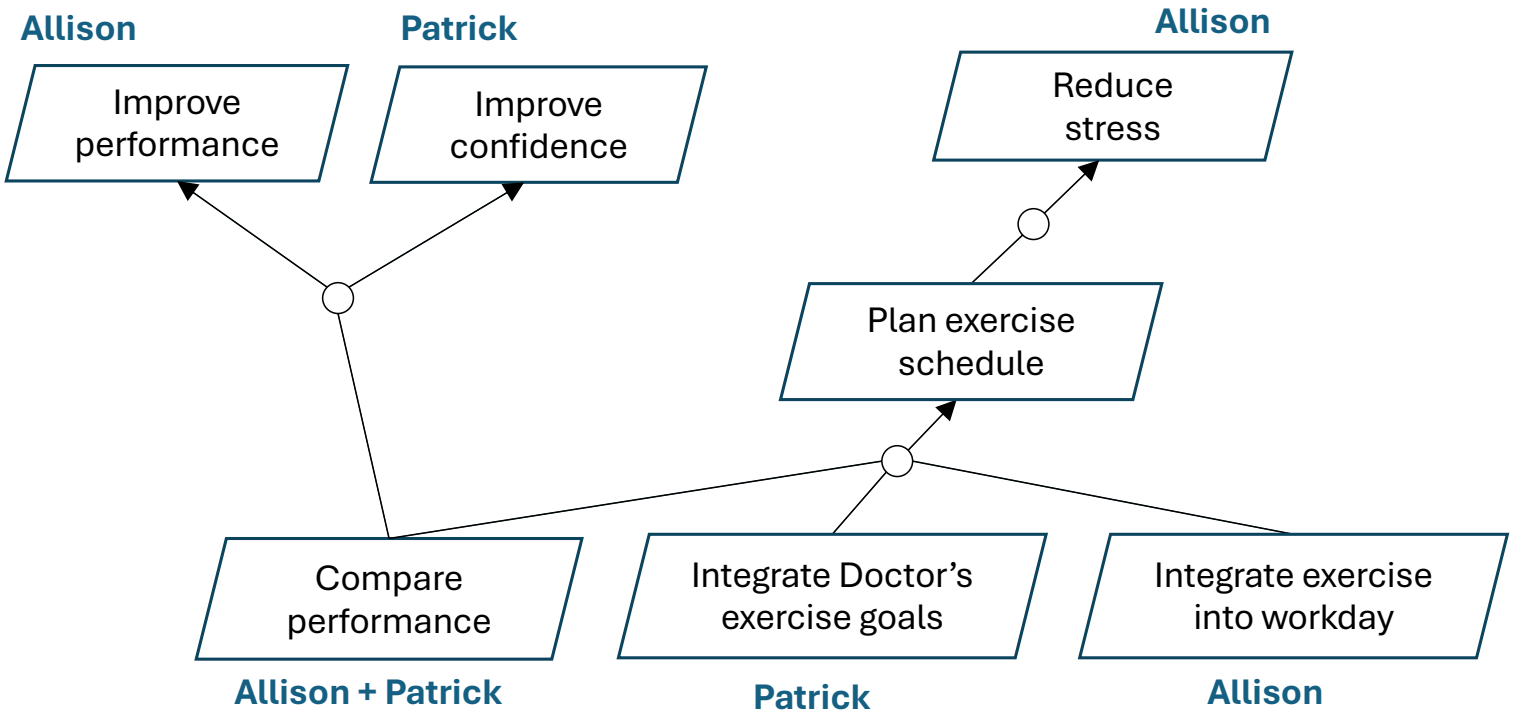


Name: Allison
Age: 59
Gender: Female
Job: Aerospace Engineer

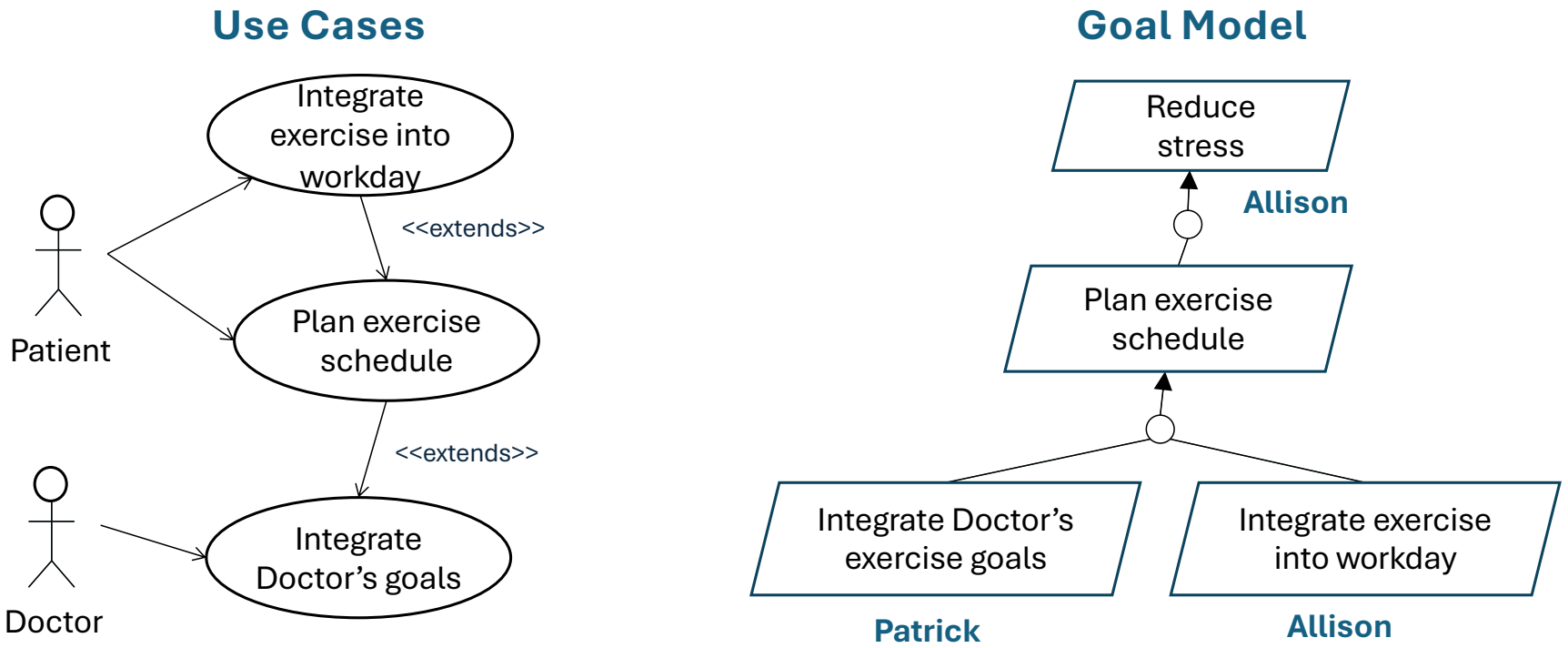


Allison changes her behavior to meet health goals, focuses on stress reduction, and is interested in modern diet options based on sound science.

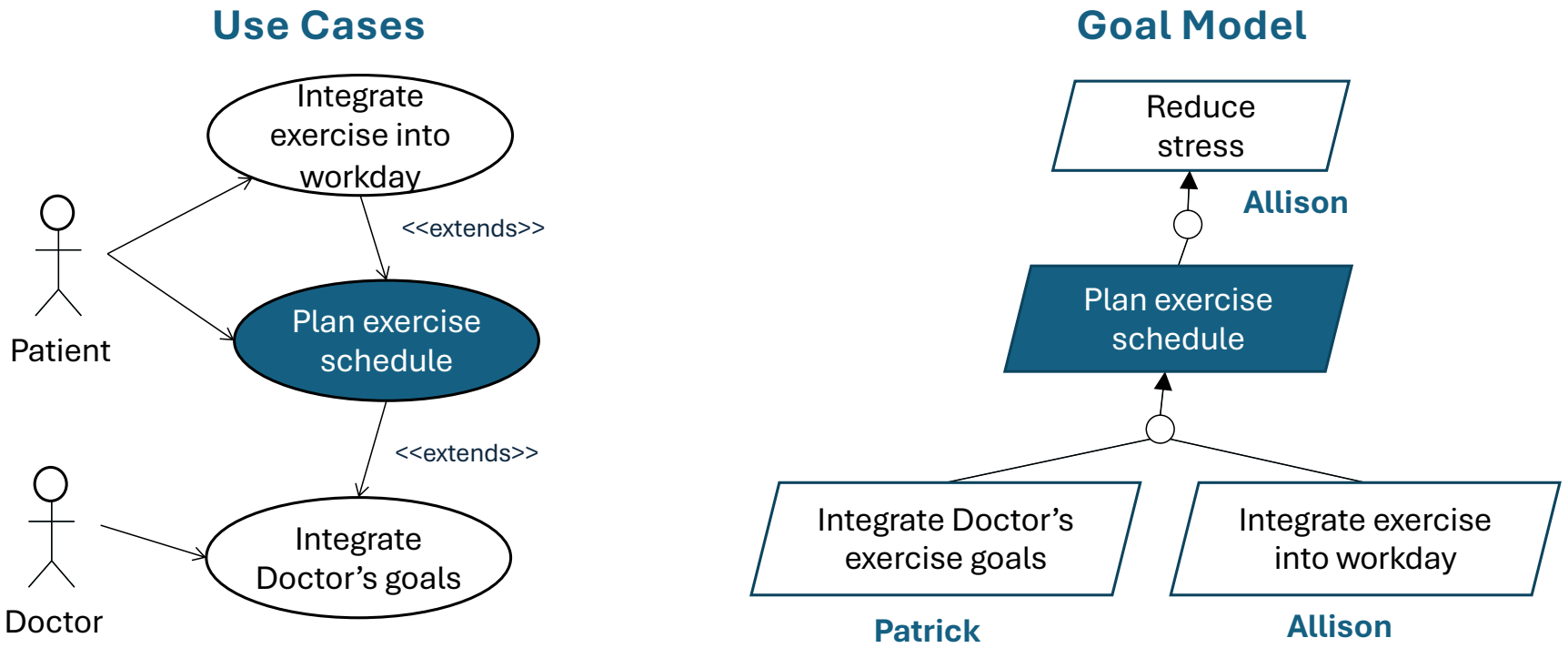
Identifying goals from personas



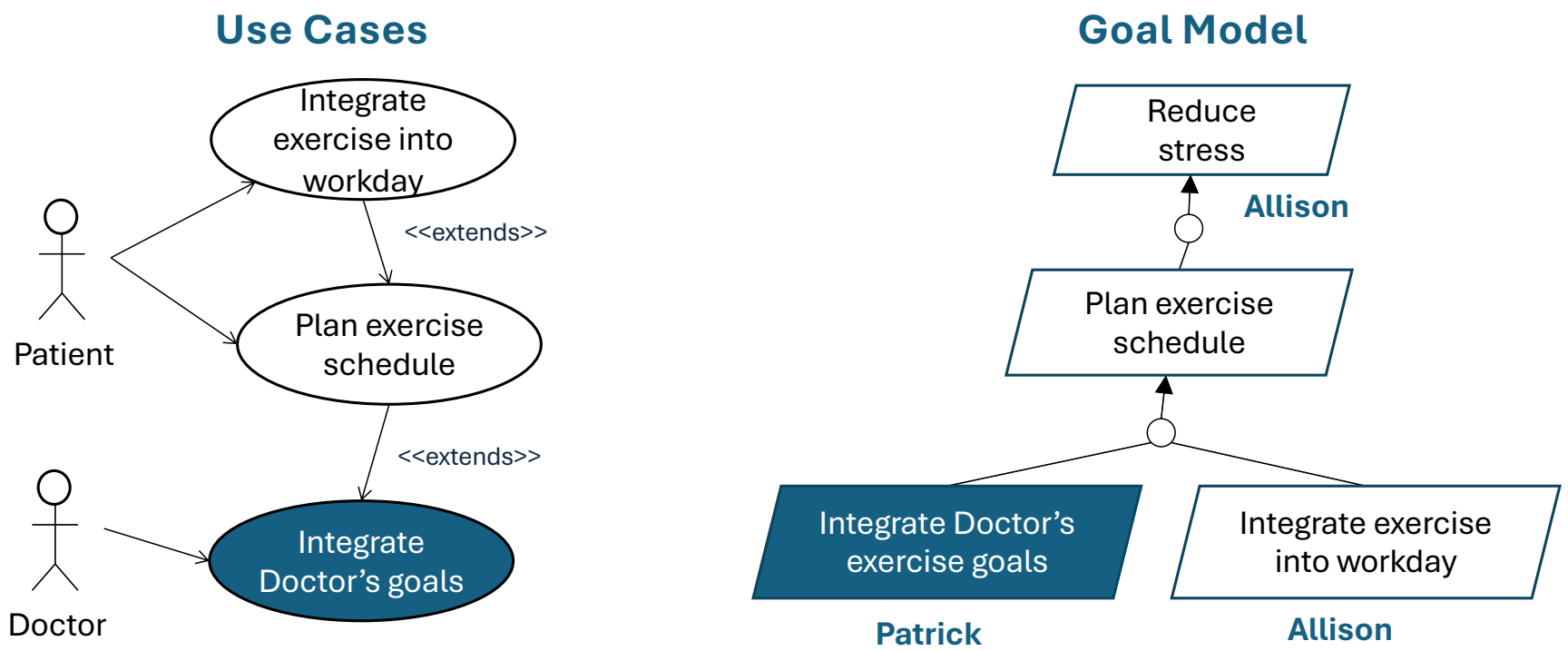
Goal model to use cases



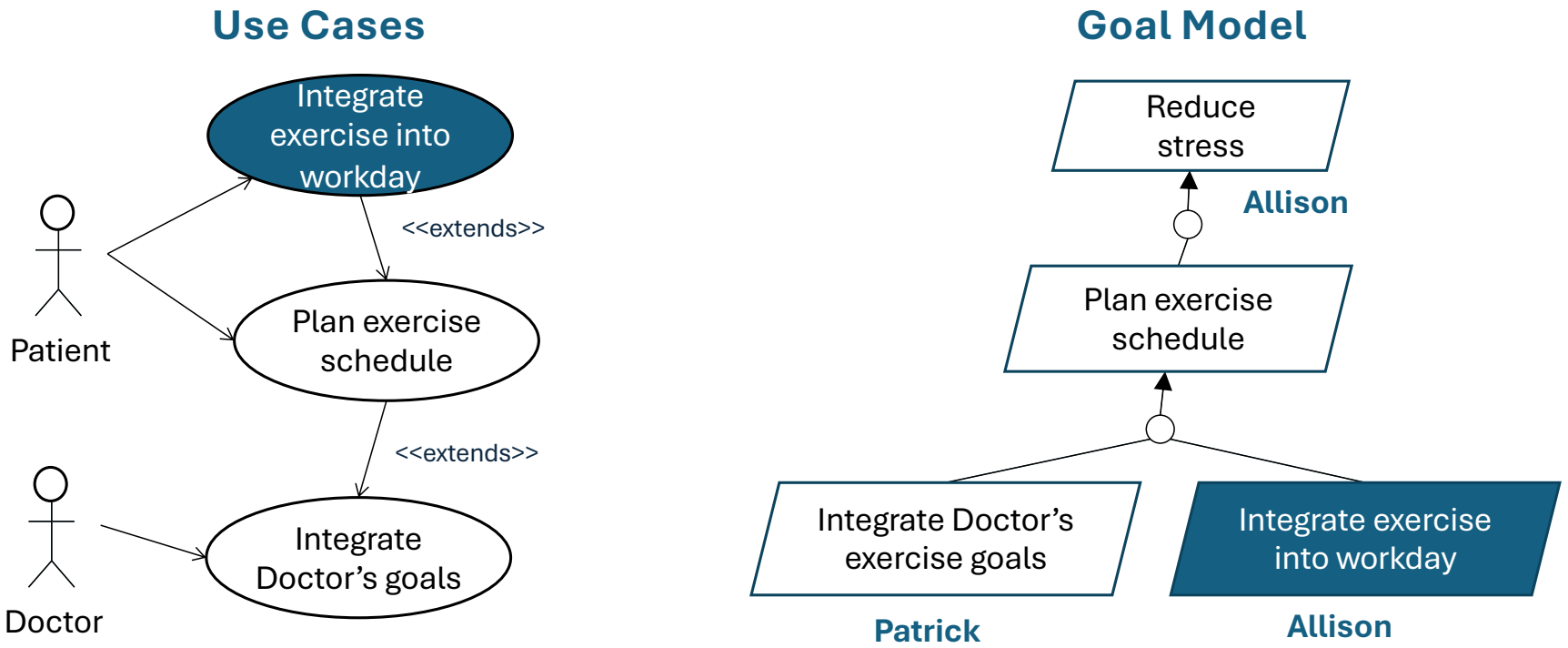
Goal model to use cases



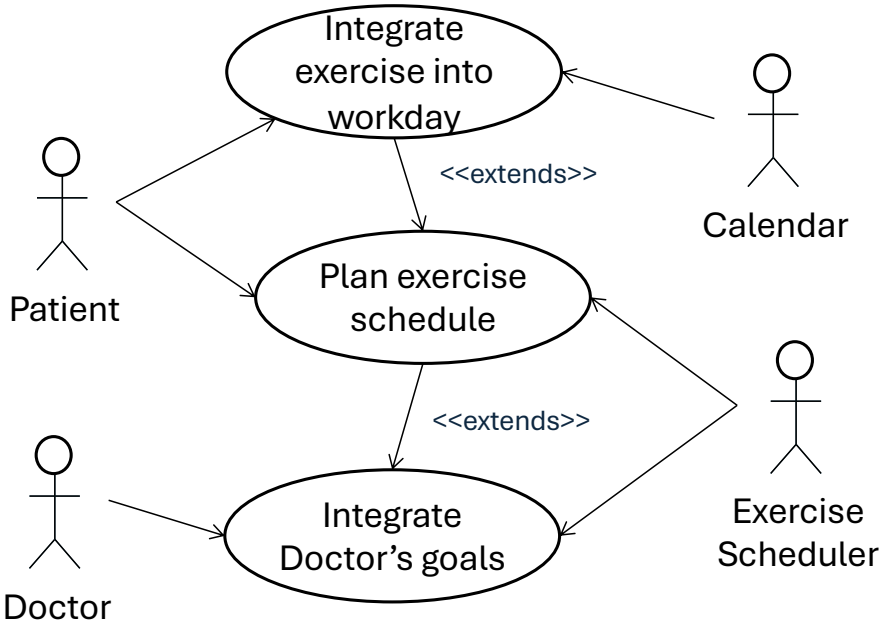
Goal model to use cases



Goal model to use cases

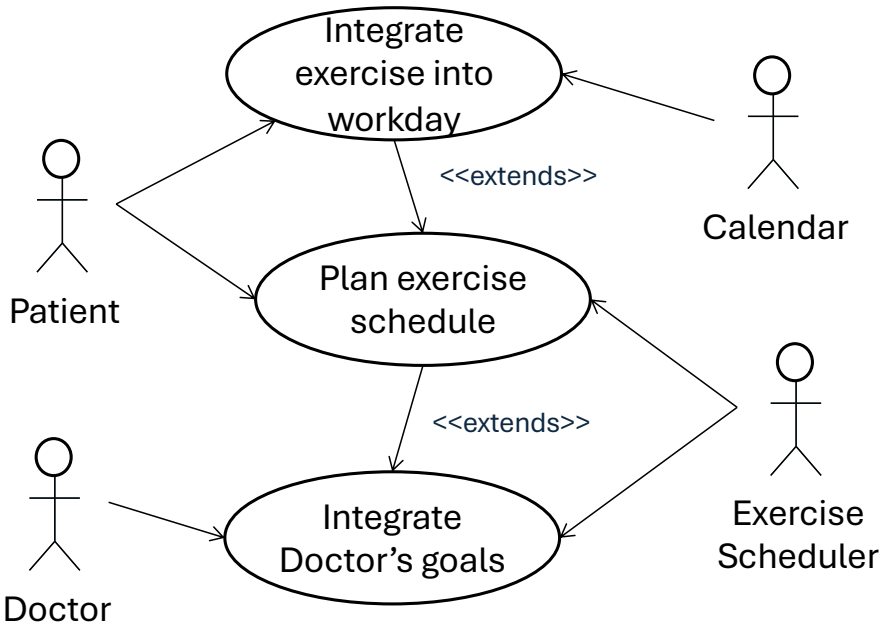


Primary v. secondary actors



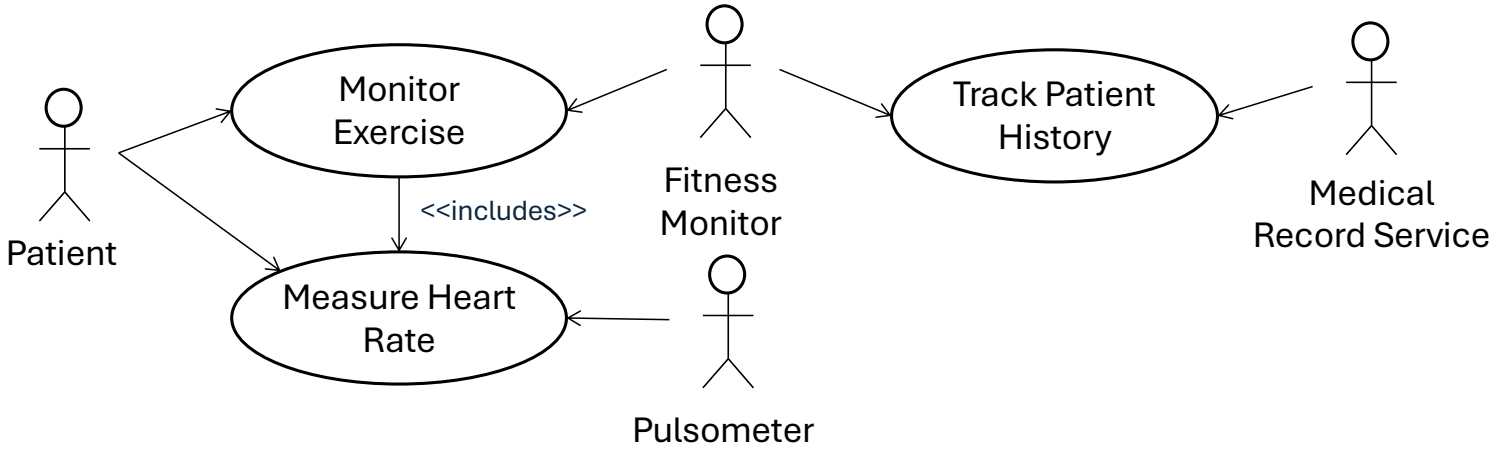
- What is a primary actor?

Primary v. secondary actors



- Primary actors *receive value from* the system; in this case, they also *initiate* the transactions
- Secondary actors *provide services to* the use case, and would not exist in the use case without the primary actors

Primary v. secondary actors, cont'd



- Actors may participate indirectly through intermediary actors
- The secondary actor in one use case can be the primary actor in a different use case

Finding use cases

Top-down: Plan exercise schedule

- Ask how questions from goal modeling to discover steps in goal statement

Bottom-up (from scenarios):

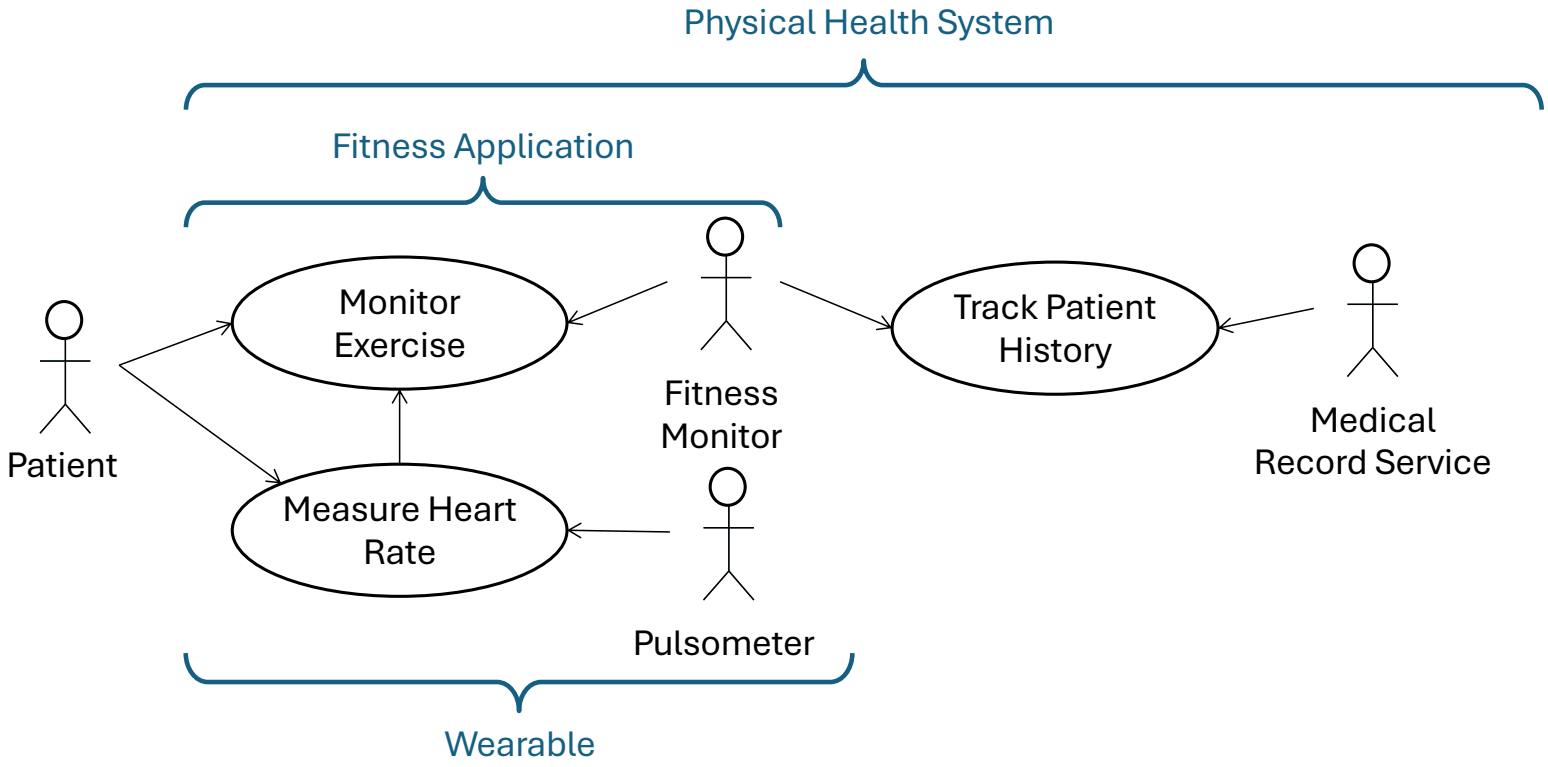
- Apply personas to create a scenario (a story) that describes how an actor would ”plan exercise schedule” in a specific setting

Example Scenario:

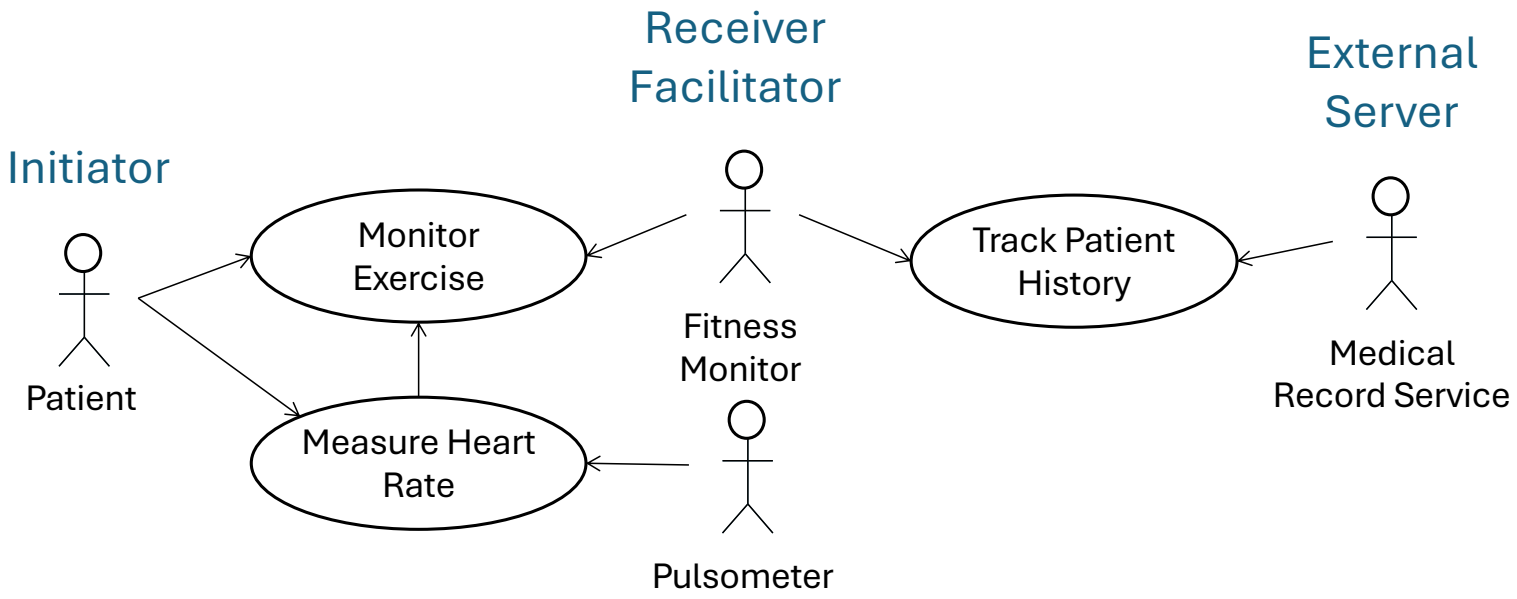
During Patrick’s annual doctor’s visit, his doctor introduces Patrick to his new fitness monitor and based on Patrick’s age, they set his first exercise goal to include a target heart rate of 75-128 bpm, and maximum heart rate of 150 bpm [1]. Next, they set his second exercise goal of 30 minutes brisk walking a day, at least five days per week.

[1] Target Heart Rate Chart, American Heart Association, January 4, 2015.

Defining the system boundaries



Actors have personalities



Use Case Templates

Use Case Name	Integrate Doctor's Goals
Actors	(Primary) Doctor, (Secondary) Patient
Pre-conditions	
Flow of events	
Post-conditions	

Pre- and post-conditions

What is the purpose of pre- and post-conditions?

- Pre-conditions: true before the use case begins
- Post-conditions: true at the end of the use case
- Should be written at the same “level of detail” as the use case
- Apply to the machine state or the observable/ controllable environment

UC: Monitor Exercise

- Pre-condition: The patient begins the exercise (walking, jumping, bending)
- Pre-condition: The patient’s resting heart rate rises above the minimum in the target range
- Post-condition: The patient’s resting heart rate falls below the minimum in the target range

Pre- and post-conditions

Example Scenario:

During Patrick's annual doctor's visit, his doctor introduces Patrick to his new fitness monitor and based on Patrick's age, they set his first exercise goal to include a target heart rate of 75-128 bpm, and maximum heart rate of 150 bpm [1]. Next, they set his second exercise goal of 30 minutes brisk walking a day, at least five days per week.

UC: Integrate Doctor's Goals

- Pre-condition: The patient has authorized the doctor to provide exercise targets
- Post-condition: The fitness frequency and/or intensity goals are stored and being monitored

[1] Target Heart Rate Chart, American Heart Association, January 4, 2015.

Use Case Templates

Use Case Name	Integrate Doctor's Goals
Actors	(Primary) Doctor, (Secondary) Patient
Pre-conditions	The patient has authorized the doctor to provide exercise targets
Flow of events	1. The doctor presents the recommended guidance based on the patient's age and mobility level 2. The patient sets the frequency goal by the number of minutes per day, and number of days per week 3. The patient sets the intensity goal using the target heart rate range and heart rate maximum
Post-conditions	The fitness frequency and/or intensity goals are stored and being monitored

Pre-conditions and assumptions

- The Doctor or Patient has an Internet connection
- The System manages user accounts
- The Doctor is familiar with the recommended guidance, or recognizes the source that publishes the guidance
- The Doctor can map the person's age and mobility level into the guidance

Which of these are domain properties, hypotheses, expectations or goals?

Use Case Name	Integrate Doctor's Goals
Actors	(Primary) Doctor, (Secondary) Patient
Pre-conditions	The patient has authorized the doctor to provide exercise targets
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Why do we need a glossary?

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Post-conditions	The fitness frequency and/or intensity goals are stored and being monitored

Example glossary

- **Recommended guidance** – guidance from the American Heart Association that describes the target heart rates by age [1]:

Age	Target Range	Maximum
20 years	100-170 beats per minute (bpm)	200 bpm
30 years	95-162 bpm	190 bpm
35 years	93-157 bpm	185 bpm
…		
70 years	75-128 bpm	150 bpm

- **Mobility level** – a physical assessment of a patient's ability to walk, run and jump based on their ability to maintain balance and stability, scored on a scale from 1 to 5

[1] Target Heart Rate Chart, American Heart Association, January 4, 2015.

Identify key system activities

- What system activities must be performed to help fulfill the use case?
  - These may not be transparent to a user

Use Case Name	Set Daily and Weekly Fitness Goals
Actors	(Primary) Fitness Monitor, (Secondary) Patient
Pre-conditions	The fitness frequency and/or intensity goals are stored and being monitored
Flow of events	1. At start of each morning, notify patient of day's goal and progress toward weekly goals 2. At start of each week, notify patient of past week's performance, and next week's goals 3. Notify patient each day when they complete their day's goals within 5 minutes of completion
Post-conditions	Patient understands their current progress toward reaching fitness goals

Alternative Flows

Alternative flows in use cases describe:

- Different processing options based on user input
- Decision taken within an existing flow
- An exception condition that occurs in a flow

Scenario and Alternate Flows

Sunny Day: Patrick completes his exercise while wearing his fitness monitor. He feels good and sees his performance improving as he's able to exercise longer at the same level of intensity. Upon completion of the exercise, his monitor sends a report to his doctor's office. The nurse reviews the report and sees no anomalies.

Rainy Day 1: … The nurse reviews the report and sees a critical anomaly, which causes her to follow up with the doctor.

Rainy Day 2: … It's a busy day in the doctor's office, and the nurse doesn't find time to review the report.

Issues?

- Patrick is feeling good, so he's unaware of the anomaly check outcome
- Anomaly checks are dependent on the nurse, and they can be busy

Scenario and Alternate Flows

Rainy Day 1: … The nurse reviews the report and sees a critical anomaly, which causes her to follow up with the Doctor.

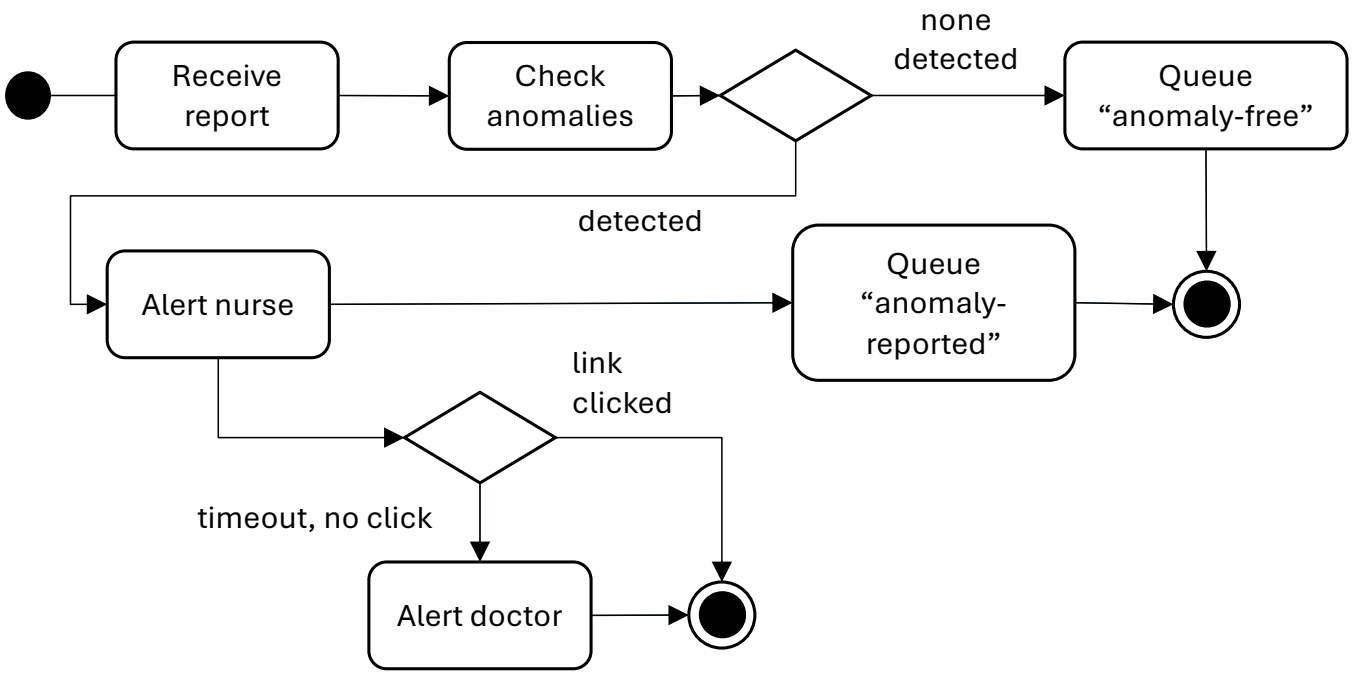
- The service performs anomaly checks and notifies the nurse and the user

Rainy Day 2: … It's a busy day in the Doctor's office, and the nurse doesn't find time to review the report.

- If the nurse does not respond within 2 hours for a critical anomaly, then the service escalates the notice to the doctor

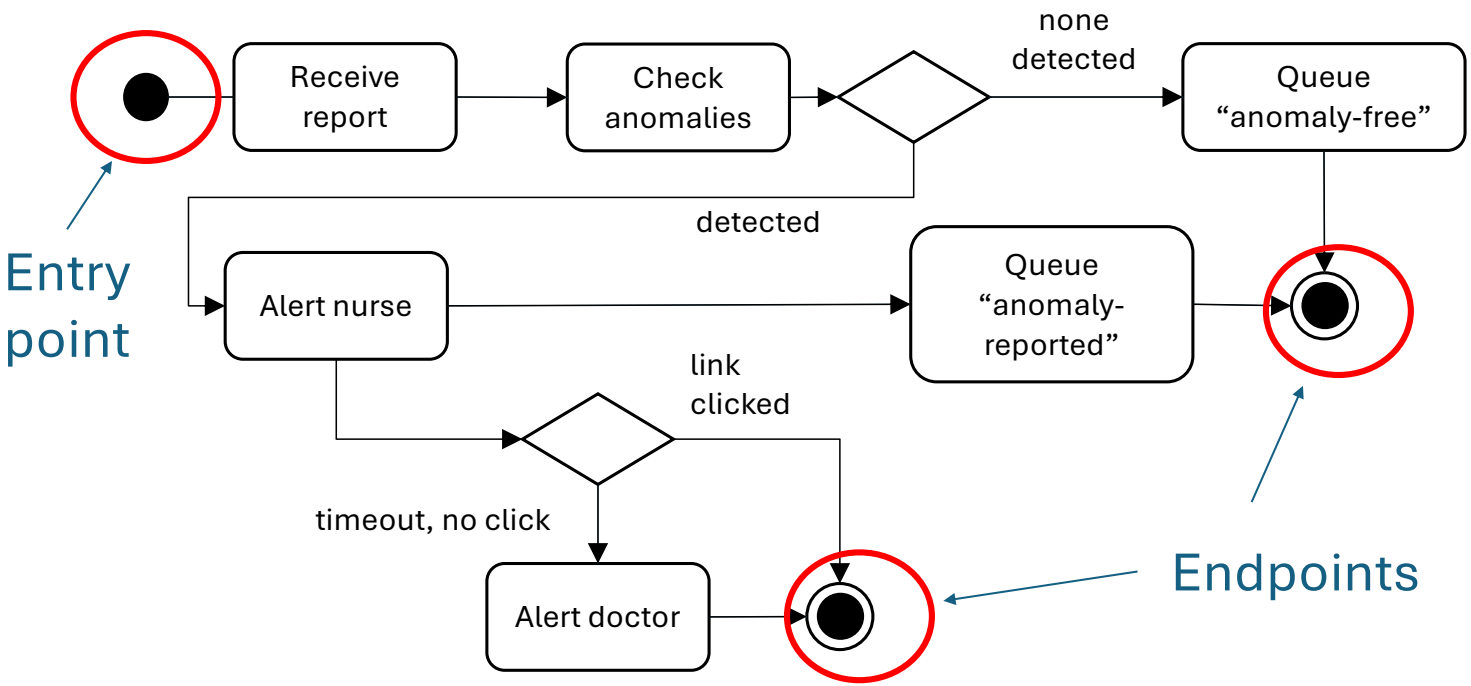
Activity Diagrams

- Activity diagrams (or flow charts) represent the conditional logic in a graph notation



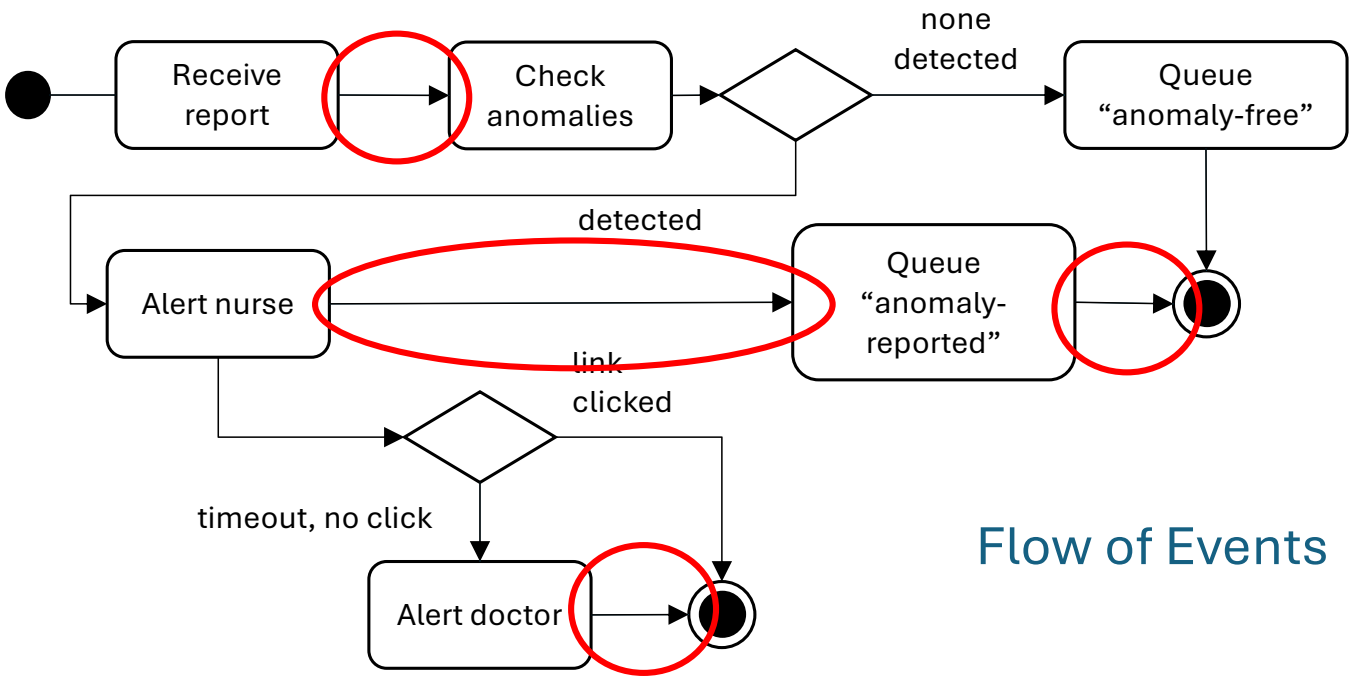
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Activity Diagrams

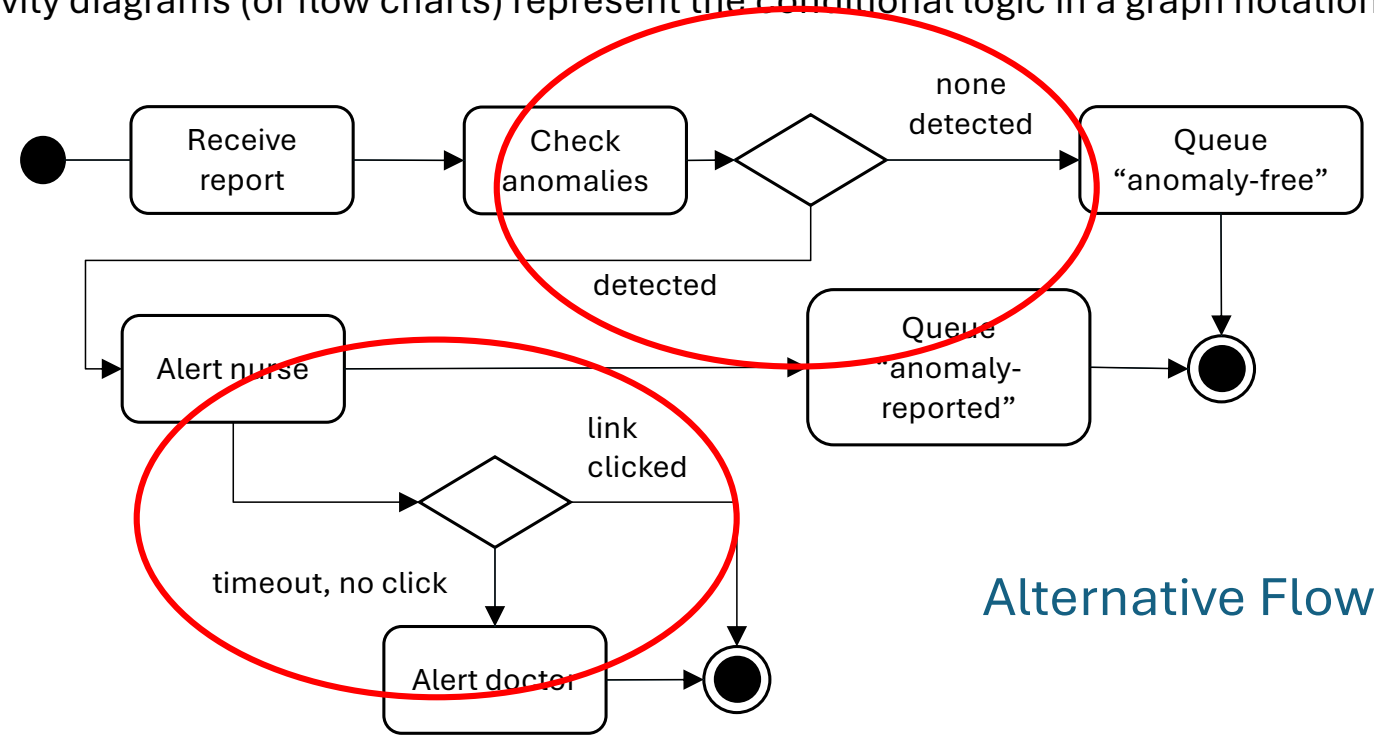
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Flow of Events

Activity Diagrams

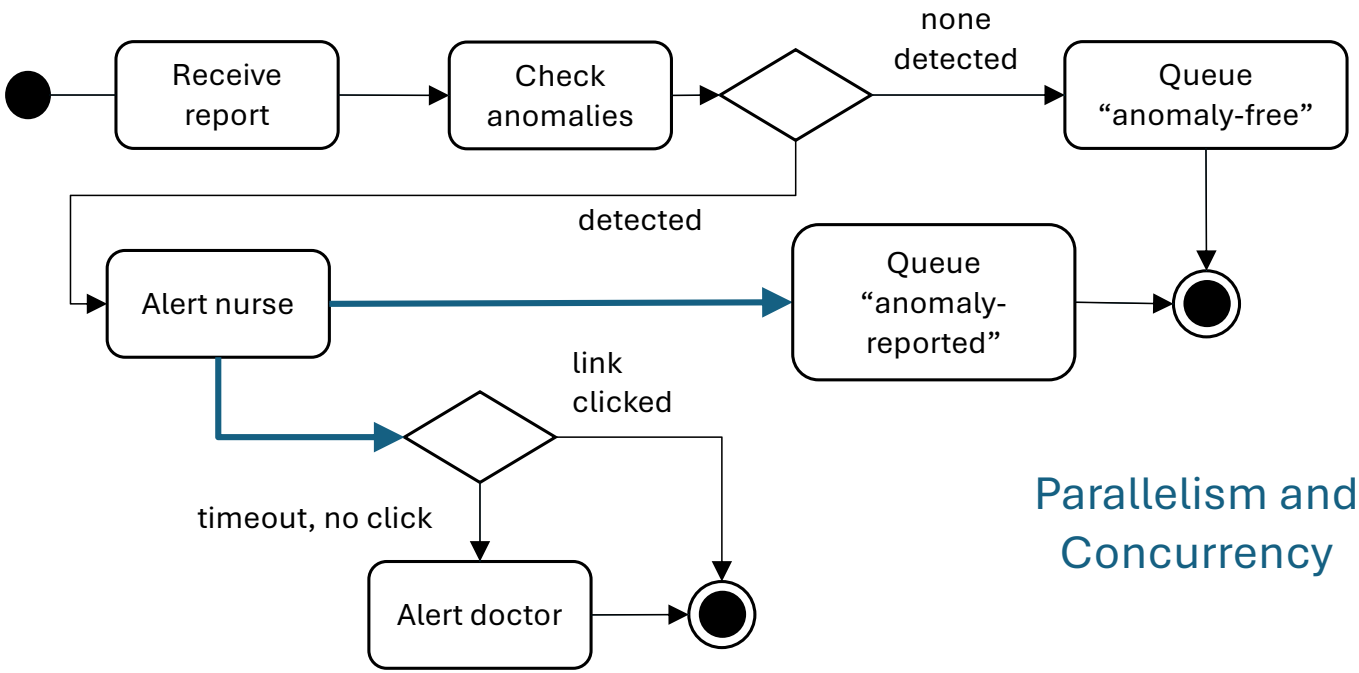
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Alternative Flows

Activity Diagrams

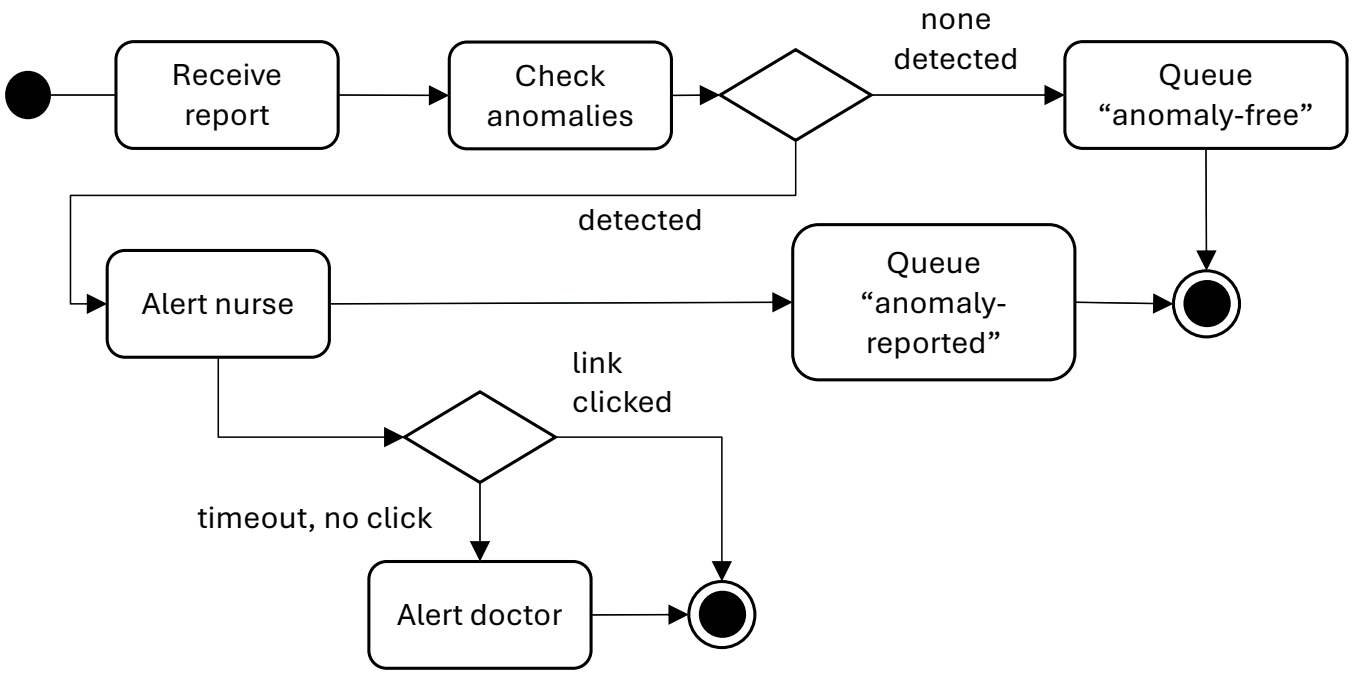
- Activity diagrams (or flow charts) represent the conditional logic in a graph notation



Parallelism and Concurrency

Activity Diagrams

- Why not just use activity diagrams? Why use use case templates?



Alternate flows & exceptions

Use Case Name	Track Patient History
Actors	(Primary) Fitness Monitor, (Secondary) Medical Record Service
Pre-conditions	The Monitor has network connectivity
Flow of events	1. The Monitor connects to the Service 2. The Monitor sends the exercise report to the Service 3. The Service checks the report for anomalies 4. The Monitor receives queued messages from Service
Post-conditions	The Service has received all exercise reports
Alternate flows and exceptions	

Alternate flows & exceptions

Use Case Name	Track Patient History
Actors	(Primary) Fitness Monitor, (Secondary) Medical Record Service
Pre-conditions	The Monitor has network connectivity
Flow of events	1. The Monitor connects to the Service 2. The Monitor sends the exercise report to the Service 3. The Service checks the report for anomalies 4. The Monitor receives queued messages from Service
Post-conditions	The Service has received all exercise reports
Alternate flows and exceptions	• The Monitor is not authorized to use the Service • The Service cannot find the Patient's record • The Service response timeout is exceeded • The Service discovers an anomaly

Integrating conditional logic

Use Case Name	Track Patient History
Actors	(Primary) Fitness Monitor, (Secondary) Medical Record Service
Pre-conditions	The Monitor has network connectivity
Flow of events	1. The Monitor connects to the Service **If** the Service is unavailable or does not respond, then… **If** the Monitor is unauthorized, then… 2. The Monitor sends the exercise report to the Service 3. The Service checks the report for anomalies **If** the Service discovers an anomaly, then… 4. The Monitor receives queued messages from Service **If** the Service is unavailable or does not respond, then…
Post-conditions	The Service has received all exercise reports

Alternate flows & exceptions

Use Case Name	Track Patient History
Actors	(Primary) Fitness Monitor, (Secondary) Medical Record Service
Pre-conditions	The Monitor has network connectivity
Flow of events	1. The Monitor connects to the Service 2. The Monitor sends the exercise report to the Service 3. The Service checks the report for anomalies 4. The Monitor receives queued messages from Service
Post-conditions	The Service has received all exercise reports
Alternate flows and exceptions	• The Monitor is not authorized to use the Service • The Service cannot find the Patient's record • The Service response timeout is exceeded • The Service discovers an anomaly

Alternate flows descriptions

Alternate Name	Unauthorized Monitor
Actors	(Primary) Medical Record Service, Fitness Monitor
Insertion Point	UC-001, Step 1, the monitor connects to the service
Pre-conditions	The report submission is initiated
Flow of events	1. The Monitor sends the secret session key 2. The Service fails to verify the session key against registered monitors 3. The Service responds that the Monitor is unauthorized
Post-conditions	The Service logs the unauthorized access request
Non-functional requirements	• Unauthorized access attempts are periodically reviewed to identify persistent threats

What do we gain by writing this alternate use case?

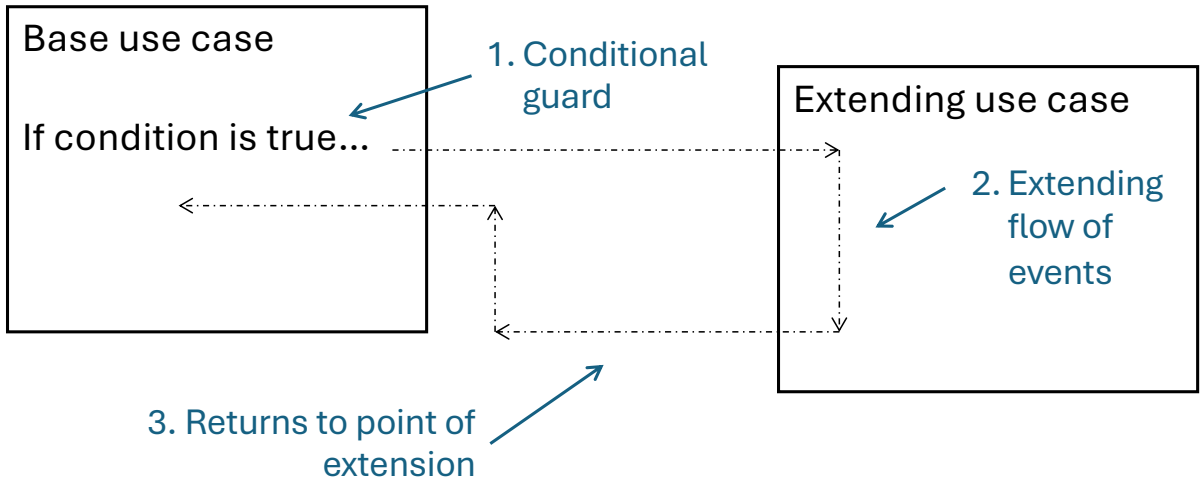
When to integrate (or not)…

- **Complex Logic**: Use an alternative flow description when the logic is complex
- **Critical Exceptions**: Use conditional logic when the exception is critical to understanding the flow
- **Multipath Flows**: If multiple paths cross through the use case, consider breaking the use case apart or using alternative flow descriptions
- **Different Priorities**: If flows use different priorities, use alternative flow descriptions

Extend, Include and Generalize

- **Extends:** model significant extensions and behaviors that can occur as additions to the base use case, but the base use case does not depend on the extends
- **Include:** model encapsulated behaviors that can be inserted into a use case and possibly reused; base use case depends on the result of the includes use case
- **Generalization**: model conceptual similarity between use cases

How do extensions work?



- If the conditional guard is true, then the extending flow is executed as part of the extending use case

Use Case Extensions

Extension Name	Anomaly Discovered
Actors	(Primary) Medical Record Service
Insertion Point	UC-001, Step 2, the service checks the report for anomalies
Pre-conditions	The heart rate exceeds the maximum rate allowed
Flow of events	1. The Service triggers an alert to the Patient's assigned nurse with a link to the exercise report 2. The Service adds the alert to the Monitor's message queue
Post-conditions	The Service has processed the anomaly
Non-functional requirements	• *What goes here?*

Use Case Extensions

Extension Name	Anomaly Discovered
Actors	(Primary) Medical Record Service
Insertion Point	UC-001, Step 2, the service checks the report for anomalies
Pre-conditions	The heart rate exceeds the maximum rate allowed
Flow of events	1. The Service triggers an alert to the Patient's assigned nurse with a link to the exercise report 2. The Service adds the alert to the Monitor's message queue
Post-conditions	The Service has processed the anomaly
Non-functional requirements	• The alert is delivered to the nurse within 60 seconds of anomaly detection • The nurse clicks the link within 2 hours of delivery • The message is received by the patient within five minutes of receiving the report

Extend v. alternate flows

- Extend use case returns flow to the base use case
  - Anomaly detected
- Alternate flows change the course of action
  - Medical Record Service is unresponsive
  - Monitor is unauthorized
- When a flow leads to a different meaning than the base use case, this is an *alternate flow*
- When a flow further defines what the base use case means, this is an *extends*

Extends v. includes

- Extend use case depends on the base
  - Conditional guard means extends is optional
- Include use case is always included, and may be shared with other base use cases

Design Pressures

- Extend use cases can simplify complex logic, similar to alternate flows
- Include use cases signal early reuse opportunities

Use case analysis

Use case analysis aims to identify defects and discover new use cases or missing elements within use cases

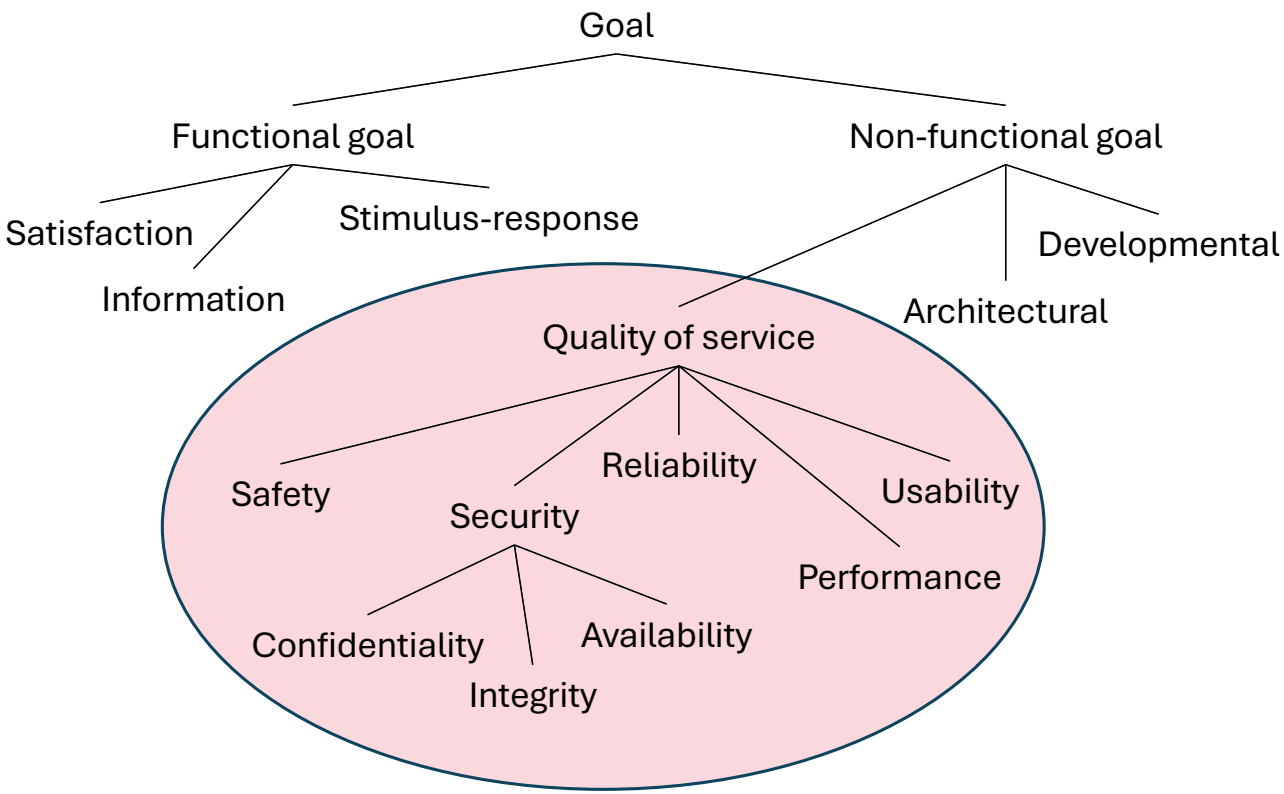
- Non-functional requirements
- Group by business functions
- Allocation and flow down
- Use case instances
- Dependency streams
- Use case prioritization

Non-functional Requirements

Non-functional requirements or quality attributes describe what stakeholders expect from the system's behavior and quality of service, for example:

- Performance: How long will the use case take to complete? What are normal and peak conditions?
- Capacity: How many actor instances must be supported?
- Security: Are there confidentiality, integrity or availability concerns?
- Usability: What do actors need to do to fulfill the use case?

Relevant goal categories



Non-functional requirements

Use Case Name	Monitor Fitness Goals
Actors	(Primary) Fitness Monitor, (Secondary) Patient
Pre-conditions	The fitness frequency and/or intensity goals are stored and being monitored
Flow of events	1. At start of each morning, notify patient of day's goal and progress toward weekly goals 2. At start of each week, notify patient of past week's performance, and next week's goals 3. Notify patient each day when they complete their day's goals within 5 minutes of completion
Post-conditions	Patient understands their current progress toward reaching fitness goals
Non-functional requirements	

Non-functional requirements

Use Case Name	Monitor Fitness Goals
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Flow of events	1. At start of each morning, notify patient of day's goal and progress toward weekly goals 2. At start of each week, notify patient of past week's performance, and next week's goals 3. Notify patient each day when they complete their day's goals within 5 minutes of completion
Post-conditions	Patient understands their current progress toward reaching fitness goals
Non-functional requirements	• Patient has opted-in to receive notifications • Notifications are timed to when patient expects to review progress toward their goals

Non-functional vs pre-conditions

- **Pre-conditions** are explicit statements about what must be true before the flow of events begins, and may be checked by the system, or otherwise assumed to be true
- **Non-functional requirements** may be satisfied by other parts of the system, but they are realized within a use case and should be accounted for

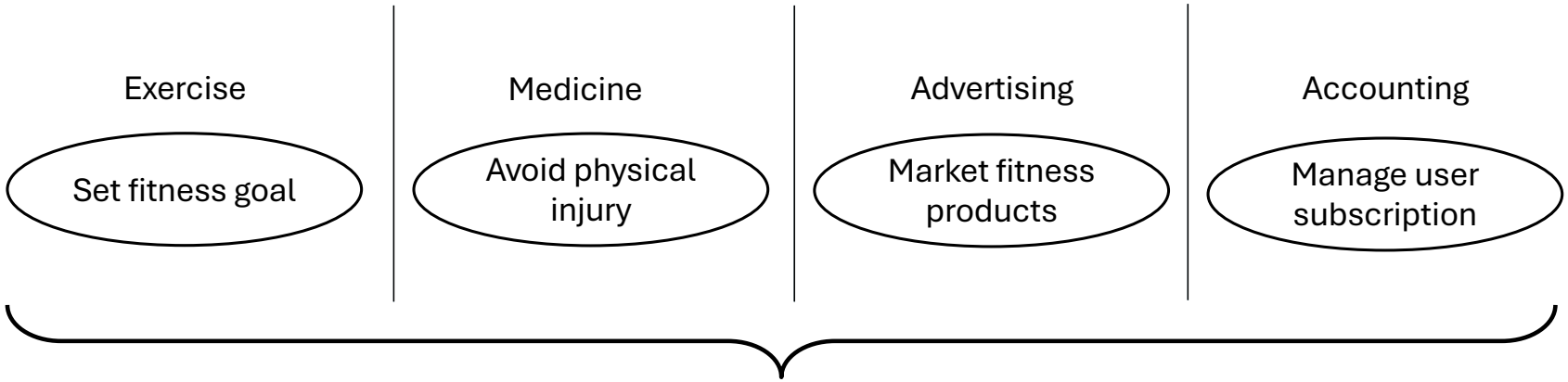
Group by business function

Business functions provide logical groups for use cases

- Exercise
  - Health Goals
  - Diet
  - Time Management
- Medicine
- Advertising
- Accounting
  - Health Plans
  - Billing
  - Payment Processing

Group by superordinate system

- Superordinate systems cut across subordinate systems, such as business functions
- What use cases derive from a fitness app when mapped across categories?

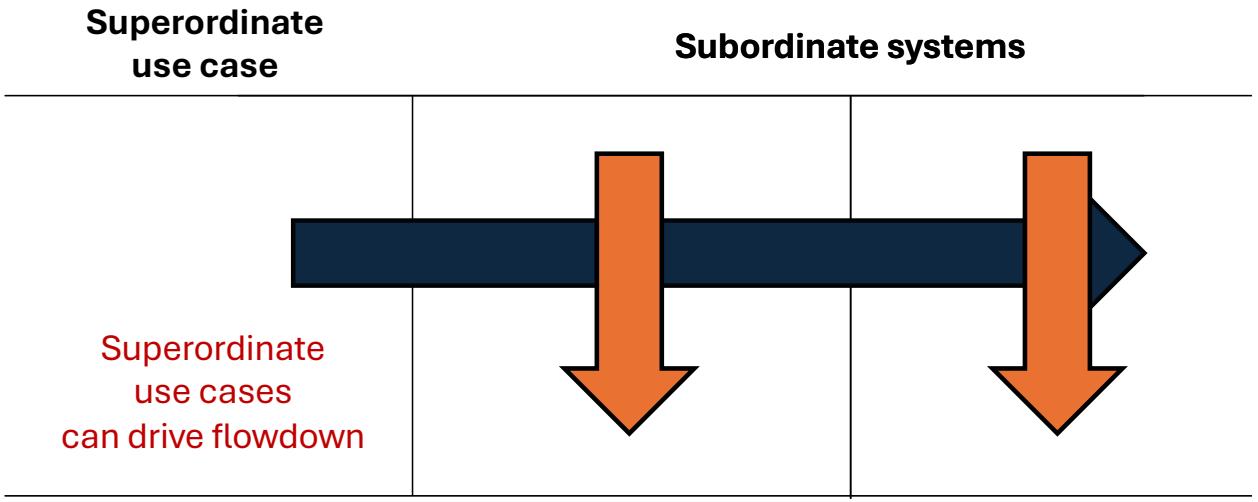


A superordinate system that traces a product or service across business functions. The analyst can now focus on developing the interfaces between these use cases

Allocation and flowdown

- Grouping by business function enables analysis
- Allocation is the assignment of requirements/use cases to subordinate systems or business functions
- Flowdown is the discovery of additional requirements from the allocated requirements

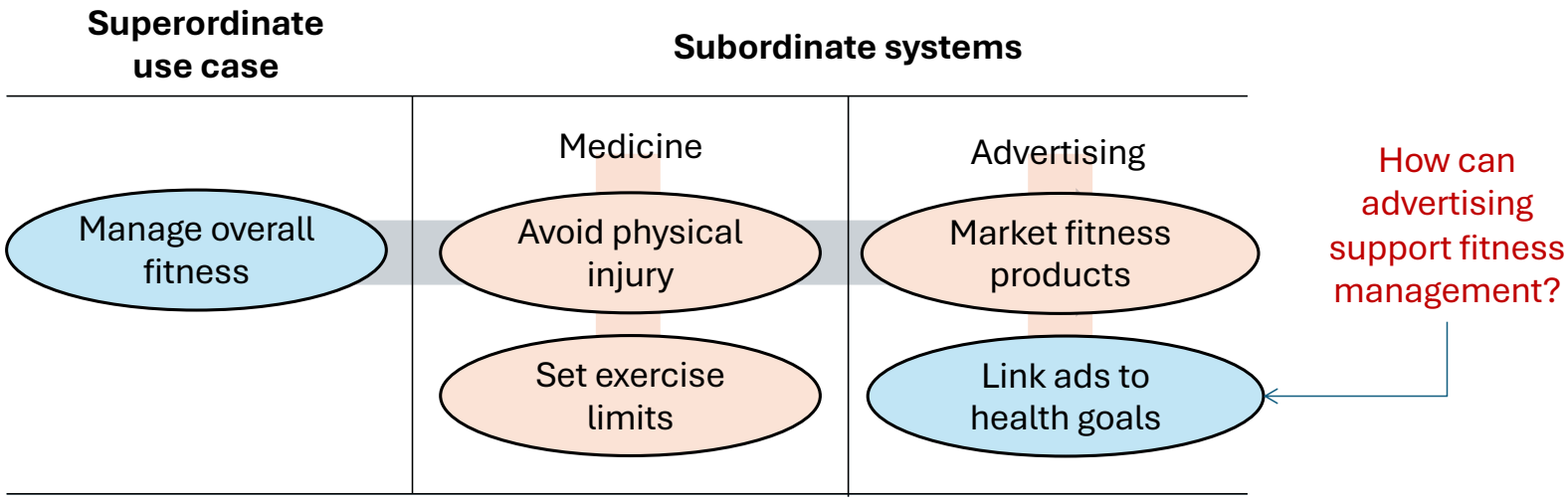
Allocation and flowdown



Allocate use cases to subordinate systems

Flowdown discovers “derived requirements” to fulfill the allocated requirements

Allocation and flowdown



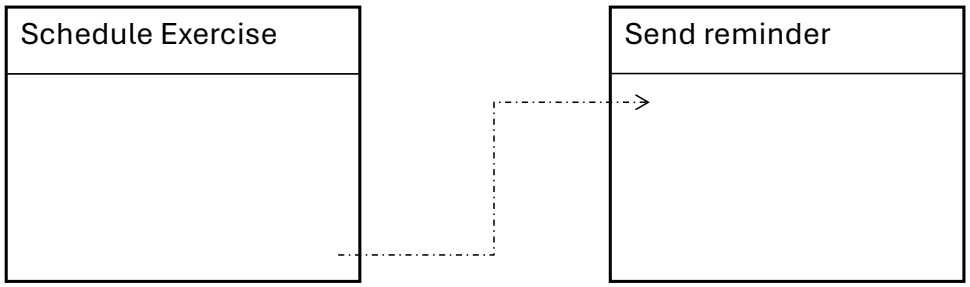
Allocate use cases to subordinate systems

Flowdown discovers “derived requirements” to fulfill the allocated requirements

Dependency Streams

- Dependency streams describe use cases that depend on each other
  - Matching pre- and post-conditions
  - Performing actions in parallel
  - Highlight interfaces between business functions

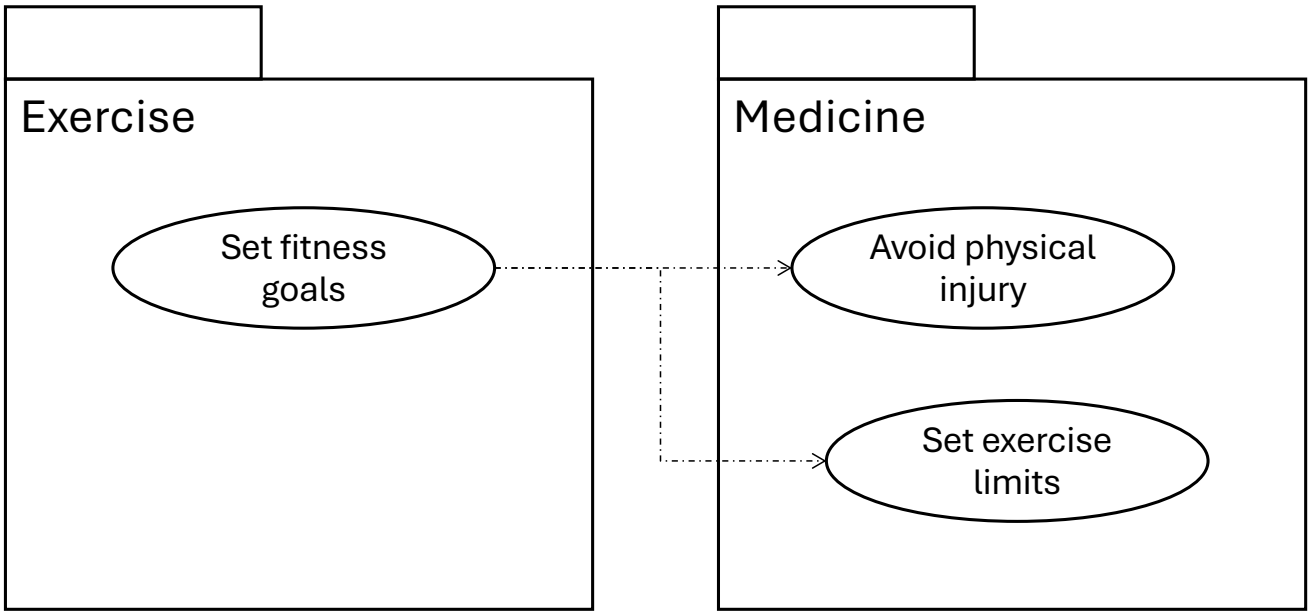
Matching pre- and post-conditions



Post-condition:
Patient has scheduled exercise

Pre-condition:
Scheduled exercise will begin within a preset number of minutes

Business functions interfaces



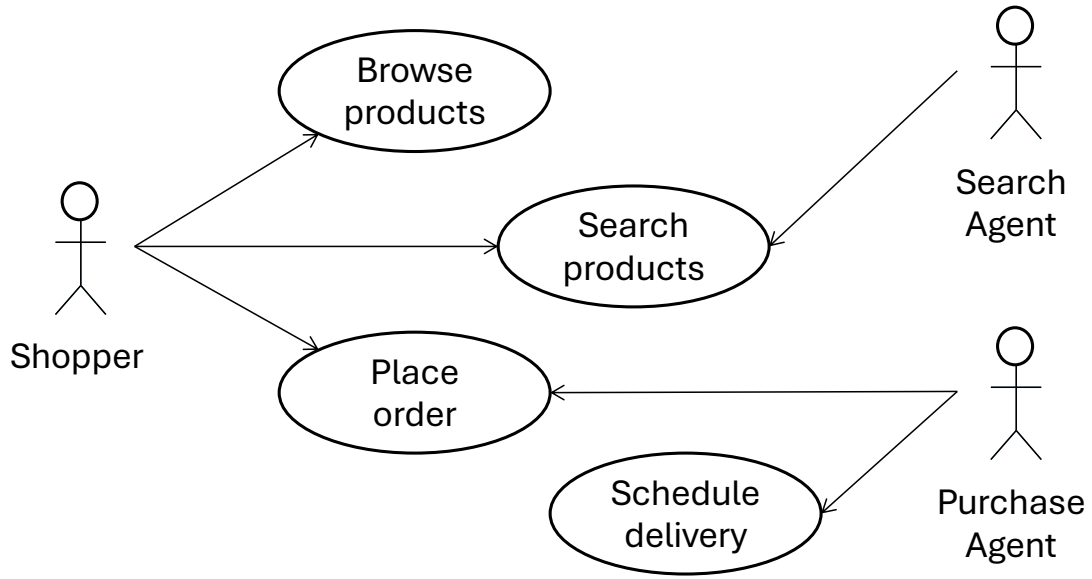
Prioritizing Use Cases

- Assigning a priority to a use case to increase or decrease the activity's importance

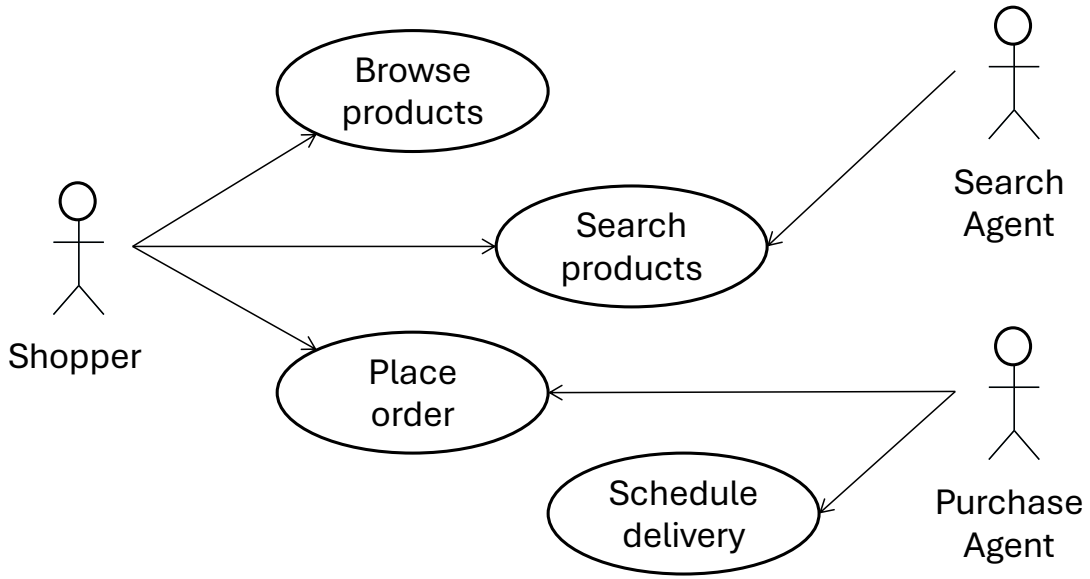
Issues to consider:

- Who sets the priority: Designer or Customer?
- Reasons for priorities: risks to quality, schedule and cost

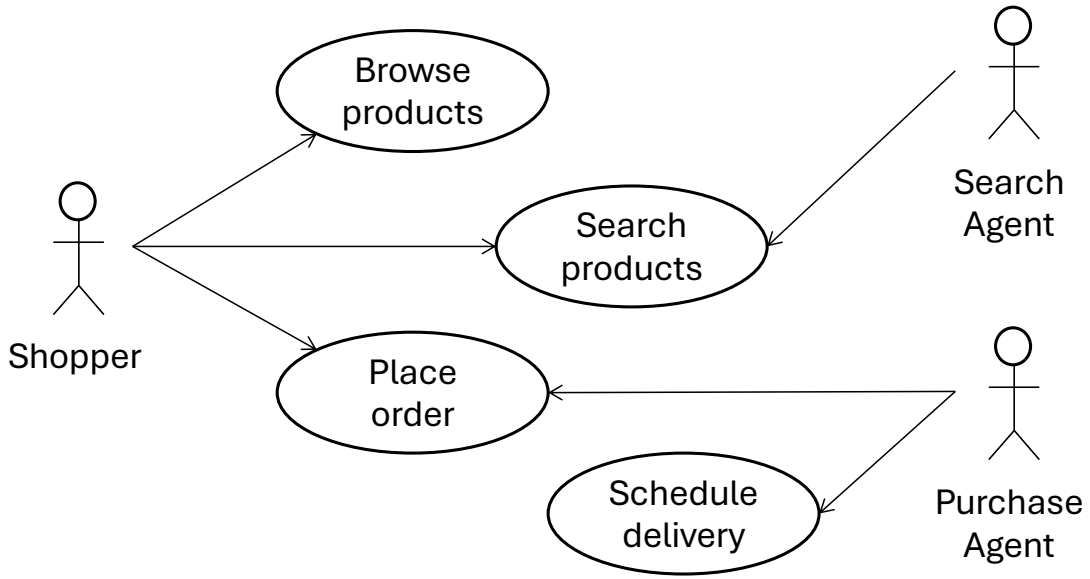
Prioritize by core functions



Prioritize by user-facing activities



Prioritize by uncertainty



Lesson Summary

- Relation of use cases to goals
- Use case diagrams
- Actors and system boundaries
- Use case templates
  - Pre- and post-conditions
  - Flow of events
  - Non-functional requirements
- Use case analysis